SUPER 60 // SKILL UP 2026

PEN-PAPER ASSESSMENT • BASICS TO LOOPS AND PATTERN PRINTING

Max Marks: 60 • Time: 2 Hours (120 Mins)

CANDIDATE NAME: _____ **ROLL NO:** _____

LAB TRACK / BATCH: _____ DATE: _____

GENERAL INSTRUCTIONS:

1. Attempt all questions. Section A (20 Marks), Section B (10 Marks), Section C (10 Marks), Section D (10 Marks), Section E (10 Marks).
2. Assume #include <iostream> and using namespace std; wherever they are not shown.
3. Show clear trace tables for output prediction and dry runs.

SECTION A: MULTIPLE CHOICE QUESTIONS

Q1. [1 Mark] What is the output of the following code?

```
int a = 7, b = 2;
cout << a / b << " " << a % b << " " << (float)a / b;
```

A) 3 1 3
B) 3 0 3.5
C) 3 1 3.5
D) 3.5 1 3.5

Q2. [1 Mark] What is the output?

```
int x = 5;
int y = x++ * 2;
cout << x << y;
```

A) 510
B) 612
C) 512
D) 610

Q3. [1 Mark] What is the output?

```
int a = 5;
cout << (a > 3 && a < 10 || a == 0);
```

A) 1
B) true
C) 0
D) Compile-time error

Q4. [1 Mark] What happens when this code is compiled and run?

```
for (int i = 0; i < 3; i++);
cout << i;
```

A) Prints 3
B) Prints 2
C) Prints 0
D) Compile-time error

Q5. [1 Mark] What is the output?

```
int i = 1;
while (i <= 5) {
    if (i == 3) continue;
    cout << i;
    i++;
}
```

A) 12345
B) 1245
C) 12 and then an infinite loop
D) 123

Q6. [1 Mark] What is the output?

```
int count = 0;
for (int i = 1; i <= 10; i++) {
    if (i % 3 == 0) continue;
    if (i == 8) break;
    count++;
}
cout << count;
```

A) 4
B) 5
C) 6
D) 7

Q7. [1 Mark] What is the output?

```
char c = 'A';
c += 2;
cout << c << (int)c;
```

(ASCII value of 'A' is 65)

A) C67 **B)** C66

C) 67C **D)** E67

Q8. [1 Mark] What is the output of cout << 5 / 2 * 2.0;?

A) 5 **B)** 2.5

C) 4 **D)** 5.0

Q9. [1 Mark] What is the output?

```
int x = 0;
if (x = 5) cout << "A";
else cout << "B";
```

A) B **B)** A

C) Compile-time error **D)** AB

Q10. [1 Mark] How many times does the inner loop body execute in total?

```
for (int i = 1; i <= 4; i++)
    for (int j = i; j <= 4; j++)
        cout << "*";
```

A) 6 **B)** 8

C) 16 **D)** 10

Q11. [1 Mark] What is the output?

```
int x = 2;
switch (x) {
    case 1: cout << "A";
    case 2: cout << "B";
    case 3: cout << "C"; break;
    default: cout << "D";
}
```

A) B **B)** BCD

C) BC **D)** C

Q12. [1 Mark] What is the output?

```
int i = 10;
do {
    cout << i;
    i++;
} while (i < 10);
```

A) Nothing is printed **B)** 10

C) Infinite loop **D)** Compile-time error

Q13. [1 Mark] What is the output?

```
int x = 7.9;
cout << x;
```

A) 8 **B)** 7

C) 7.9 **D)** Compile-time error

Q14. [1 Mark] What is the output?

```
int a = 5;
cout << !a + 1;
```

A) 0 **B)** 2

C) 1 **D)** 6

Q15. [1 Mark] What is the value of x?

```
int x = 2 + 3 * 4 % 5 - 1;
```

A) 3 **B)** 1

C) 4 **D)** 14

Q16. [1 Mark] Which of the following code fragments results in an infinite loop?

A) for (int i = 10; i > 0; i--) { }
B) int i = 0; while (i < 5); i++;
C) int i = 0; while (i < 5) i++;
D) do { } while (0);

Q17. [1 Mark] What pattern does this code print?

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= 3 - i; j++) cout << " ";
   for (int j = 1; j <= i; j++) cout << "* ";
   cout << endl;
}
```

A) A centred pyramid
B) A left-aligned right triangle
C) An inverted triangle
D) A rectangle

Q18. [1 Mark] What is the output?

```
int n = 1234, r = 0;
while (n) {
   r = r * 10 + n % 10;
   n /= 10;
}
cout << r;
```

A) 1234
B) 10
C) 4321
D) 0

Q19. [1 Mark] What is the output?

```
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      if (j == 1) break;
      cout << i << j << " ";
   }
}
```

A) 00 01
B) 00 10 20
C) 00 01 02
D) 00

Q20. [1 Mark] What is the output?

```
int x = 5;
cout << (x > 3 ? x < 10 ? "A" : "B" : "C");
```

A) B
B) C
C) 1
D) A

SECTION B: ERROR FINDING

Q21. [5 Marks] (5 Marks) The program should read n and print the sum of the first n natural numbers. Identify each of the 5 errors by line number, explain what is wrong, and write the correction.

```
1  #include <iostream>
2  using namespace std
3  int Main()
4  {
5      int n, sum = 0;
6      cout >> "Enter a number: ";
7      cin >> n;
8      for (int i = 1, i <= n, i++)
9          sum += i;
10     cout << "Sum = " << Sum << endl;
11     return 0;
12 }
```

Q22. [5 Marks] (5 Marks) The program should read n, print its factorial, print the digits 0 to 4, and report if n is zero. Identify 5 errors by line number, explain what is wrong, and write the correction.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter n: ";
7      cin >> n;
8      int fact;
9      for (int i = 1; i <= n; i++) {
10         fact = fact * i;
11     }
12     cout << "Factorial = " << fact << endl;
13     int i = 0;
```

SECTION C: DEBUGGING

Q23. [10 Marks] (10 Marks) The program below is meant to read number of rows n and print a centred star pyramid followed by total stars printed (for n = 4, expected 16 stars).

```
#include <iostream>
using namespace std;

int main() {
    int n;
    int stars;
    cout << "Enter number of rows: ";
    cin >> n;

    for (int i = 0; i <= n; i++) {
        for (int s = 1; s <= i; s++) {
            cout << " ";
        }
        for (int j = 1; j <= i; j++) {
            cout << "*";
            stars++;
            cout << endl;
        }
    }
    cout << "Total stars printed: " << stars;
    return 0;
}
```

(a) Identify the 5 bugs and explain why each is wrong. (5 marks)
(b) Write the complete corrected program. (3 marks)
(c) Dry run for n = 3 and write the exact output. (2 marks)

SECTION D: OUTPUT PREDICTION AND DRY RUN

Q24. [5 Marks] (5 Marks) What is the exact output of the program?

```
#include <iostream>
using namespace std;

int main() {
    int n = 4;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (j <= n - i) cout << " ";
            else cout << j - (n - i);
        }
        cout << endl;
    }
    return 0;
}
```

(a) Write the exact output, keeping the alignment. (4 marks)
(b) How many times is cout << " " executed in total? (1 mark)

Q25. [5 Marks] (5 Marks) Write the exact output of the program.

```
#include <iostream>
using namespace std;

int main() {
    int total = 0;
    for (int i = 1; i <= 10; i++) {
        if (i % 2 == 0) continue;
        if (i > 7) break;
        for (int j = 1; j <= i; j++) {
            if (j % 3 == 0) total += j;
            else total -= 1;
        }
        cout << i << ":" << total << " ";
    }
    cout << endl << "Final: " << total;
    return 0;
}
```

SECTION E: PROGRAMMING

Q26. [5 Marks] (5 Marks) Write a complete C++ program that reads n and prints the following number pyramid (sample for n = 5). Use only loops and conditions (no arrays or strings).

```
    1
   121
  12321
 1234321
123454321
```

Q27. [5 Marks] (5 Marks) Write a complete C++ program that reads a positive integer and prints (a) the number of digits, (b) the sum of its digits, and (c) whether the number is a palindrome. For example, for input 1221 output should be: Digits: 4, Sum of digits: 6 and Palindrome. Use only loops and conditions (no arrays or strings).